



Abdulrahman Munir

AI/ML Specialist — Hausa Speech-to-Text (ASR) & Transcription

Profile

AI/ML specialist and developer focused on **low-resource natural language processing and automatic speech recognition (ASR)**. Native **Hausa** speaker who grew up in Hausa-speaking communities and also learned **Arabic** from childhood. Combining machine learning engineering skills with genuine linguistic fluency to build **speech-to-text and transcription systems for Hausa**, a language historically underserved by commercial ASR tools.

Core Strengths

Speech & Audio	Automatic Speech Recognition (ASR), fine-tuning Whisper and wav2vec2/XLS-R, forced alignment, VAD and audio denoising, audio dataset curation, transcription and annotation
NLP	Low-resource NLP, language modelling, tokenization, transliteration (Arabic script ajami & Latin boko for Hausa), corpus building
Linguistic Edge	Native speaker of Hausa, childhood education and fluency in Arabic, strong command of English; deep familiarity with Nigerian accents and dialectal variation across the Hausa-speaking region
ML Engineering	PyTorch, TensorFlow/Keras, Hugging Face Transformers & Datasets, scikit-learn, pandas/NumPy, experiment tracking
Dev & MLOps	Python, Git, Docker, AWS (S3/EC2), Linux, Jupyter, Label Studio (data annotation), ffmpeg/audio tooling

Technical Skills

Languages	Python, Bash, (learning) C++	Frameworks	PyTorch, TensorFlow, Hugging Face
Speech/ASR	Whisper, wav2vec2/XLS-R, Librosa, Torchaudio	NLP	NLTK/Hugging Face NLP, tokenizers, low-resource pipelines
Data	pandas, NumPy, Datasets, Label Studio	Audio/Dev	ffmpeg, Sox, Docker, Git, AWS, Linux

Projects

- 2024–present **Hausa Speech-to-Text Transcription System**, *Independent project, Python, Whisper, Hugging Face, PyTorch*
- Designed and developed an end-to-end **Hausa ASR pipeline**: audio ingestion, VAD/denoising, transcription, and timestamped output in both Latin (*boko*) and Arabic (*ajami*) orthographies.
 - Fine-tuned **OpenAI Whisper** and **wav2vec2-XLSR** on Mozilla **Common Voice** Hausa data plus self-collected recordings.
 - Curated and cleaned a custom **Hausa audio dataset** and applied **forced alignment** and quality filtering to lift transcription accuracy on conversational speech.
 - Evaluated low-resource techniques (augmentation, orthographic normalisation, language-model rescoring) to close the gap on out-of-vocabulary terms.

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- 2024 **Hausa–Arabic Transliteration & Script Tool**, *Independent project*, **Python**, **rule-based** + **neural mapping**
- Built a tool converting between Hausa **boko** (Latin) and **ajami** (Arabic-script) writing systems, leveraging childhood Arabic literacy.
 - Designed a phonetic mapping table plus a small alignment model to support transcription and linguistic search.

Experience

- 2025 **Image Annotation for Object Detection**, *Independent / client-contracted work*, Nigeria
- Annotated object bounding boxes in images using **LabelImg**, exported in Pascal VOC (XML) and YOLO formats for model training.
 - Applied consistent label conventions; reviewed and corrected annotations to minimise mislabelled boxes and keep data quality high.
 - Produced a clean, labelled dataset that fed directly into an object-detection model pipeline.

Education

- In progress **B.Eng. Computer Engineering**, *Federal University of Technology, Minna*, Nigeria
- Relevant coursework: Machine Learning, Deep Learning, NLP, Data Structures, Statistics.

Languages

English	Professional proficiency (working language)	<i>Read, write, speak</i>
Hausa	Native (fluent, incl. dialectal variants)	<i>Read, write, speak</i>
Arabic	Native-level from childhood education	<i>Read, write, speak</i>

Personal Projects

- 2026 **Hausa Spoken-Audio Dataset Building**, *Personal*, **ffmpeg**, **Python**
- Recording, cleaning and organising Hausa speech samples for personal speech-to-text experiments, exercising audio preprocessing knowledge from the ASR pipeline.
- 2024–present **Athena — Voice Assistant with Speech-to-Text**, *Personal (voice/AI assistant)*, **Python**, **React**, **STT/TTS**
- Building a local-first AI voice assistant (Python backend, React UI) with on-device **speech-to-text** transcription, text-to-speech and a "*Hey Athena*" wake-word demo.
- 2024–present **Qur'an Recitation Video Generator**, *Personal*, **Python**, **ffmpeg**, **SRT timing**
- Python generator rendering recitation videos with **word-by-word Arabic subtitles synced to audio** (SRT), plus a timing editor aligning transcript segments to exact audio timestamps.
- 2024–present **ÌròyìnCheck — Multi-Platform News-Checking Bot**, *Personal*, **Node.js**, **Baileys**, **WhatsApp/Telegram/Discord**
- Fact-checking/news-checking bot delivered across **WhatsApp (Baileys v7)**, **Telegram** and **Discord** from a single resilient Node.js codebase.
- 2024–present **Linux Automation & Utility Scripts**, *Personal (home projects)*, **Bash**, **Python**, **Linux**
- Authoring personal Bash/Python scripts that automate repetitive file, audio and media workflows on a Linux desktop, saving time on everyday tasks.